# Dimensional Modeling Concepts – A Foundation for Data Warehousing

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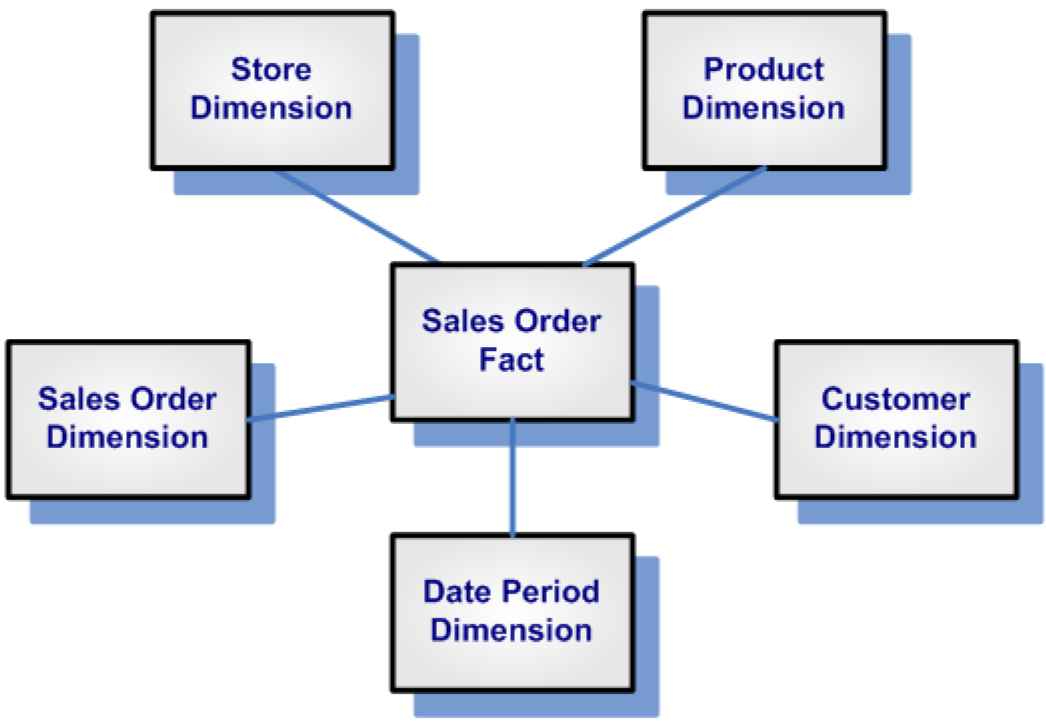
## 1. Slide: What is Dimensional Modeling?

**Description:**
Dimensional modeling is a data design technique optimized for data retrieval, primarily used in data warehouses. It simplifies data for end-user queries by organizing it into **facts** and **dimensions**.

**Illustration:**

- A visual: cube symbolizing OLAP operations (slice, dice, roll-up).




**Code Sample:**

```
-- Example structure for sales fact
CREATE TABLE fact_sales (
  sale_id INT,
  customer_id INT,
```

```
  product_id INT,
  store_id INT,
  sales_amount DECIMAL(10,2),
  sale_date DATE
);
```

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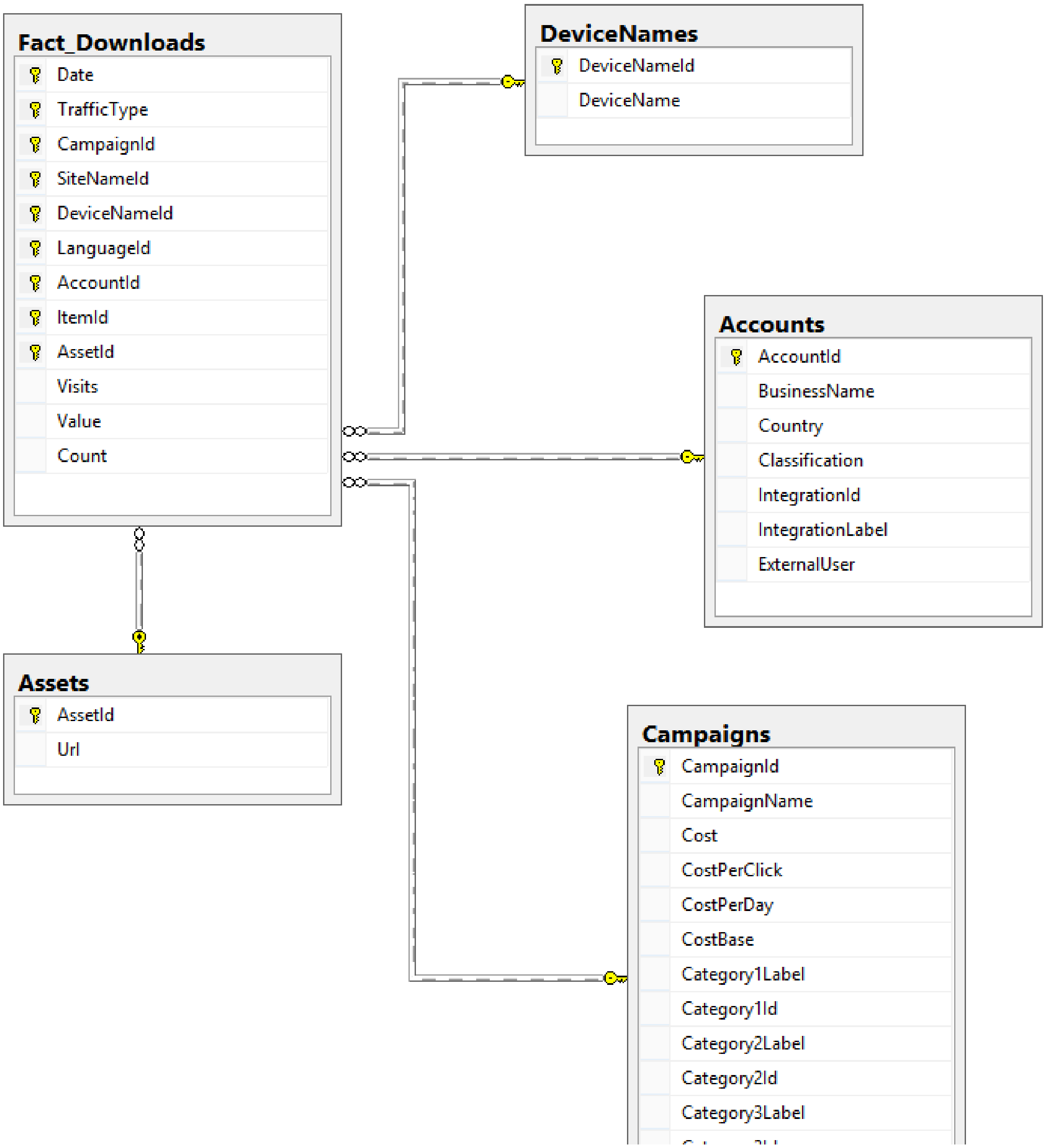
## 2. Slide: Fact and Dimension Tables

**Description:**

- **Fact Tables:** Contain measurable, quantitative data.
- **Dimension Tables:** Contain descriptive attributes to describe the facts.

**Illustration:**

- Star schema showing a fact table in the center surrounded by dimension tables.




**Code Sample:**

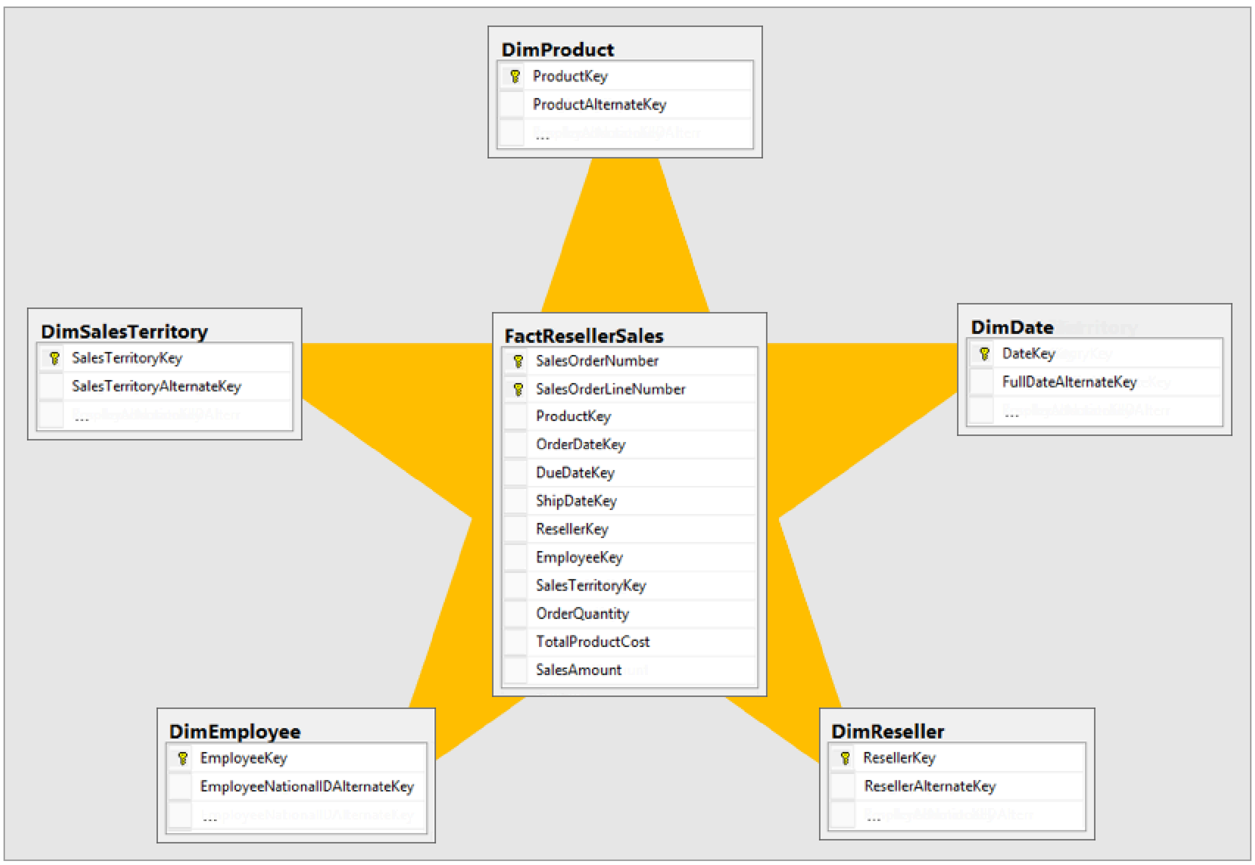
```
-- Dimension: Customer
CREATE TABLE dim_customer (
  customer_id INT PRIMARY KEY,
  name VARCHAR(100),
  region VARCHAR(50)
);
```

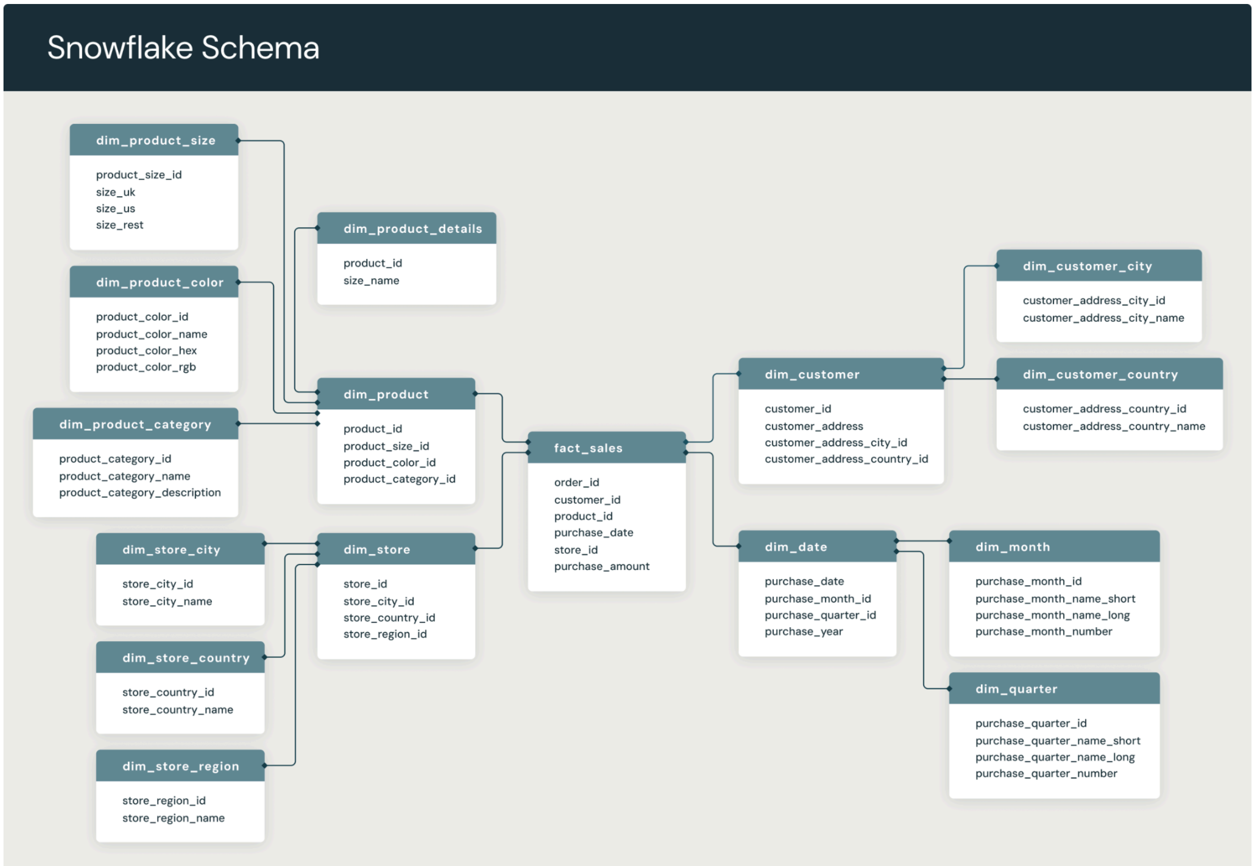
## 3. Slide: Database Schema

**Description:**
Schemas define how tables are logically organized. In data warehouses, schemas are designed for fast aggregation, typically using dimensional modeling.

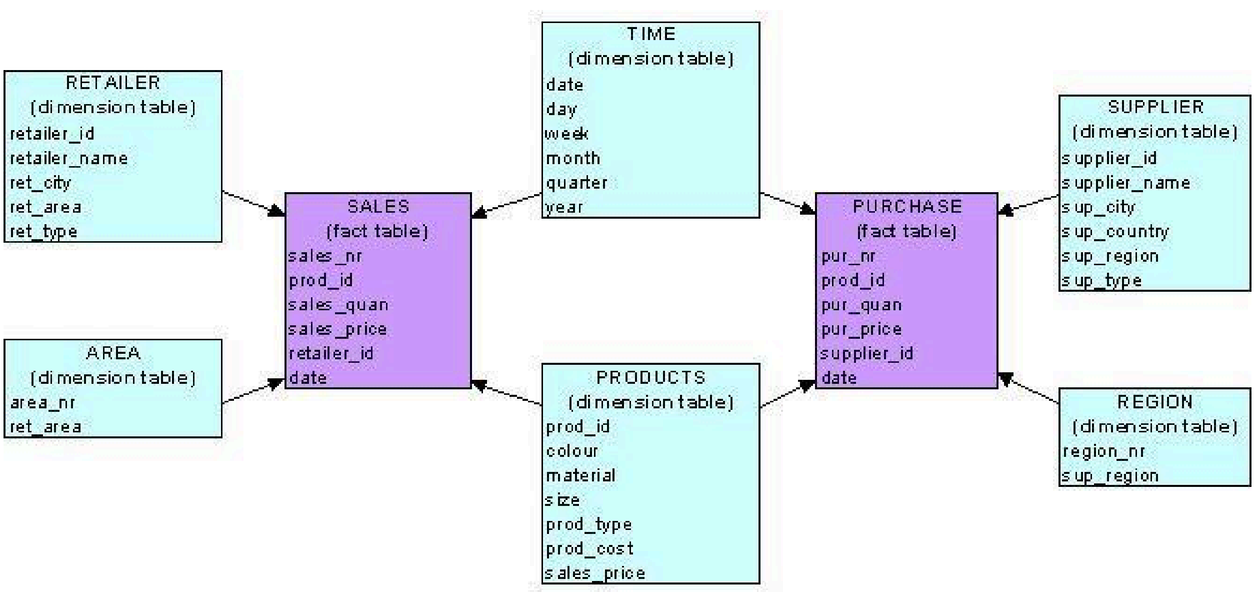
**Key Types:**

- Star Schema
- Snowflake Schema
- Fact Constellation Schema






Galaxy Model



## 4. Slide: Schema Design for Modeling

**Description:**
Schema design involves:

- Identifying facts
- Defining granularity
- Identifying dimensions
- Choosing schema type (star, snowflake, constellation)

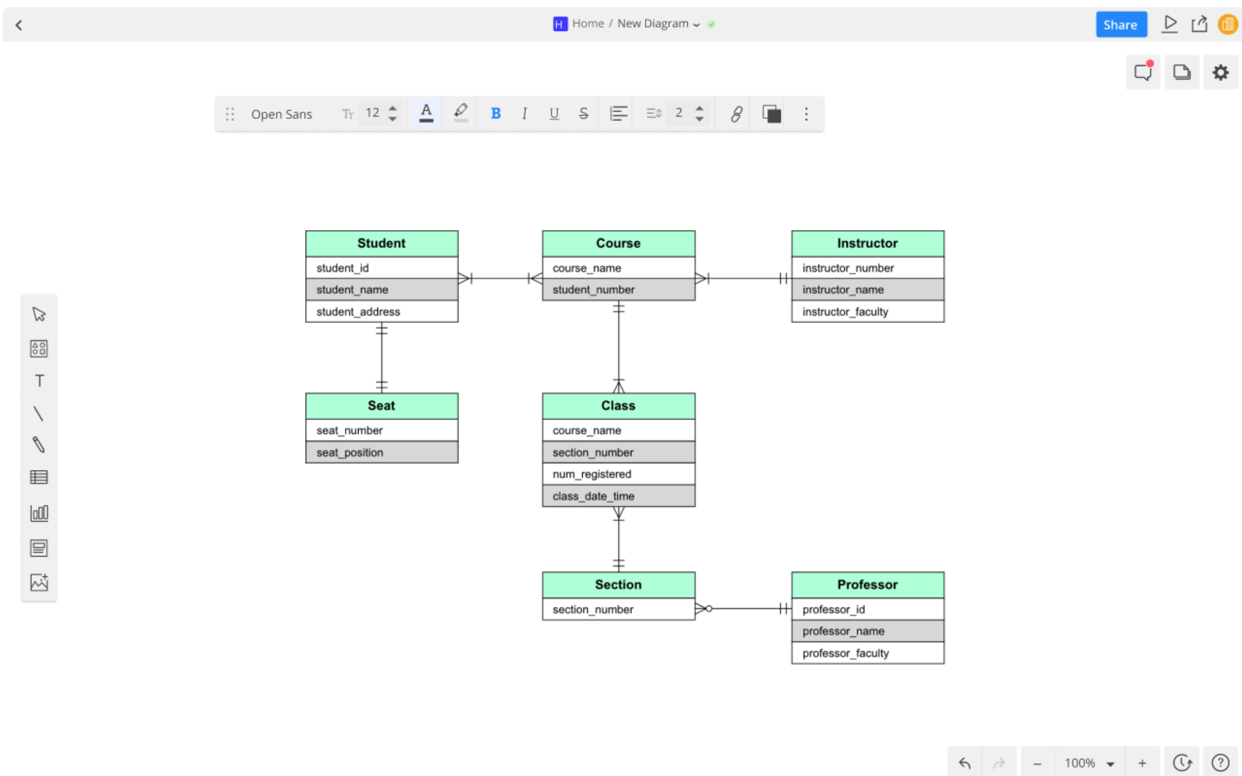



**Illustration:**
Diagram comparing normalized OLTP vs. denormalized OLAP (DW) schema.

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## 5. Slide: Star Schema

**Description:**
In a Star Schema:

- Fact table in center.
- Dimension tables connected directly (denormalized).
- Simple, fast querying.

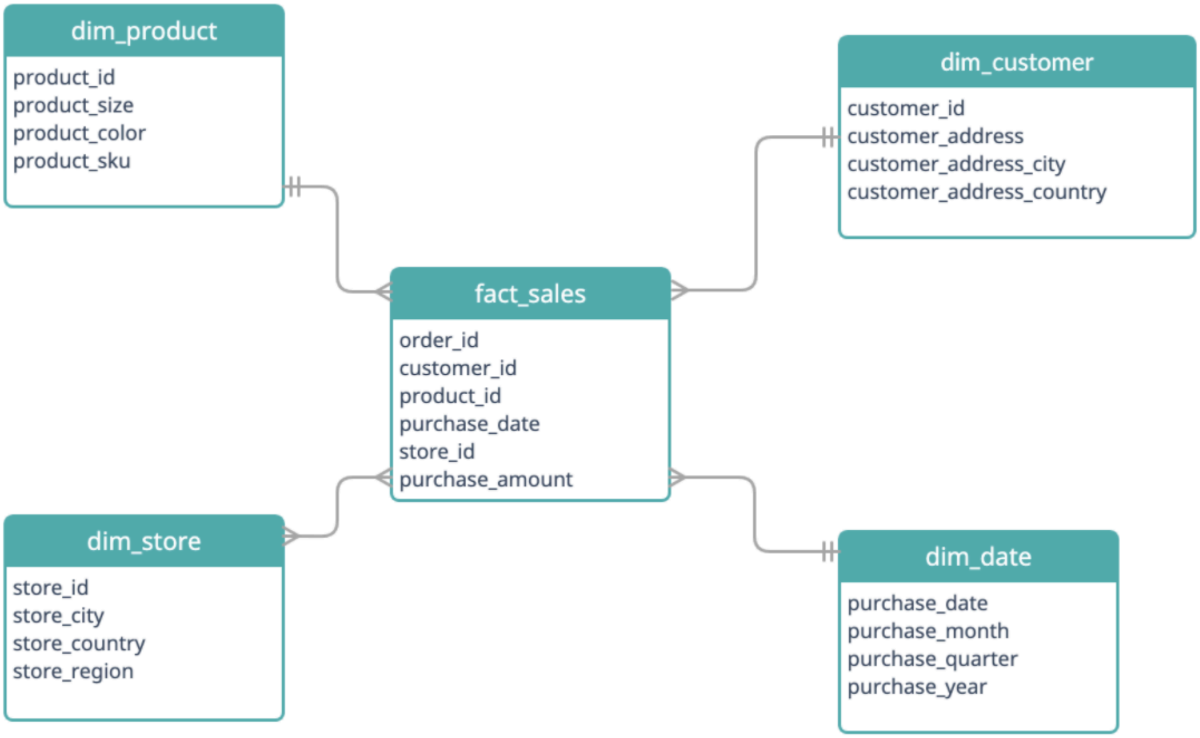


**Illustration:**
Star schema diagram with `fact_sales` and surrounding `dim_customer`, `dim_product`, `dim_time`.

**Code Sample:**

```
SELECT c.name, p.category, f.sales_amount
FROM fact_sales f
JOIN dim_customer c ON f.customer_id = c.customer_id
JOIN dim_product p ON f.product_id = p.product_id;
```

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## 6. Slide: Snowflake Schema

**Description:**

- Normalized dimension tables (sub-dimensions).
- More joins, slower but saves space.
- Useful for hierarchical data.

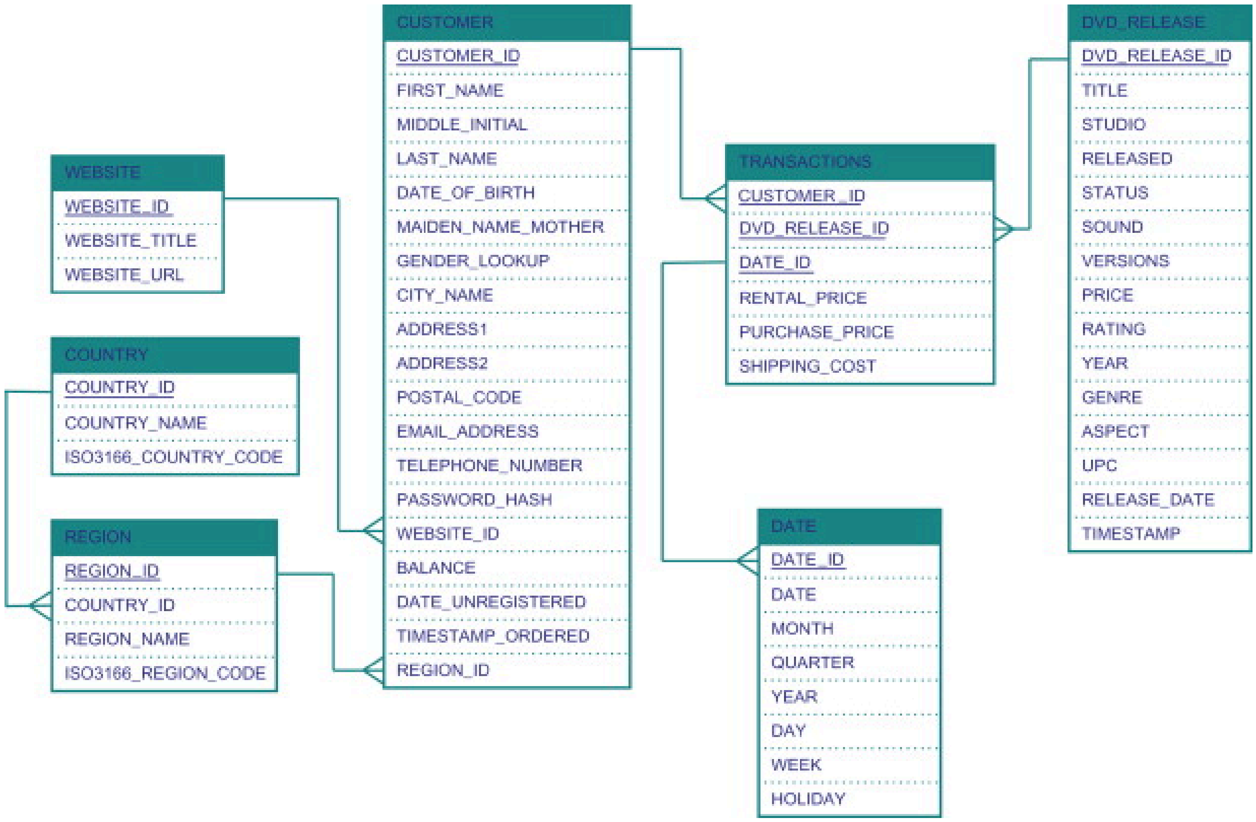


**Illustration:**
Snowflake schema with normalized `dim_location` → `dim_country`, `dim_state`.

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## 7. Slide: Fact Constellation Schema

**Description:**
Also known as **Galaxy Schema**:

- Multiple fact tables share dimension tables.
- Useful for complex BI systems.

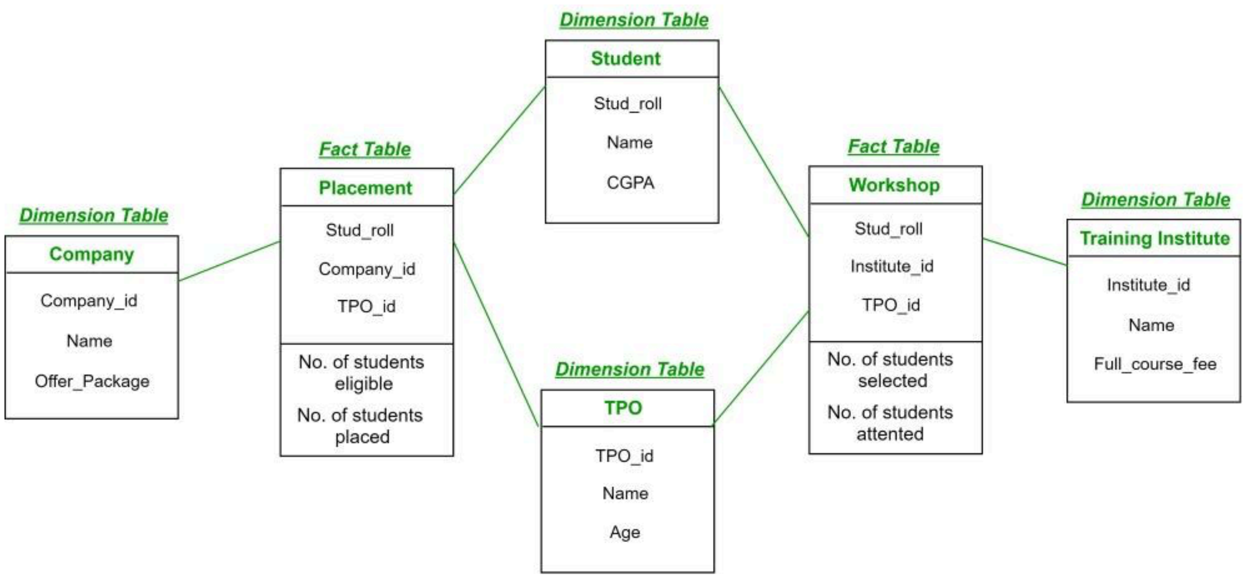


**Illustration:**
Diagram showing `fact_sales` and `fact_returns` sharing dimensions like `dim_time`, `dim_customer`.

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## 8. Slide: Key Comparison Table

| Schema Type | Simplicity | Query Speed | Storage | Use Case |
|---|---|---|---|---|
| Star | High | Fast | Moderate | Simple, fast-reporting systems |
| Snowflake | Medium | Medium | Low | Normalized, space-efficient |
| Fact Constellation | Complex | Variable | Variable | Complex analytics, shared dims |

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## 9. Slide: Summary of Dimensional Modeling

- Organizes data for analytics.
- Uses fact and dimension tables.
- Star and Snowflake schemas are popular.
- Galaxy schema supports multi-fact scenarios.

## 🔧 Optional Additions

**Tools/Use Cases:**

- Tools: Power BI, Tableau, SSAS
- Use Cases: Retail sales analysis, Healthcare KPIs, Banking fraud detection

**SQL Analytics Sample:**

```sql
-- Top 5 customers by total sales
SELECT c.name, SUM(f.sales_amount) AS total_sales
FROM fact_sales f
JOIN dim_customer c ON f.customer_id = c.customer_id
GROUP BY c.name
ORDER BY total_sales DESC
LIMIT 5;
```

## 💼 Next Steps

- Demo schema in PostgreSQL or SQL Server
- Excel dashboard with pivot table on sample sales data
- Optional: Connect to Power BI/Tableau for a live demo